

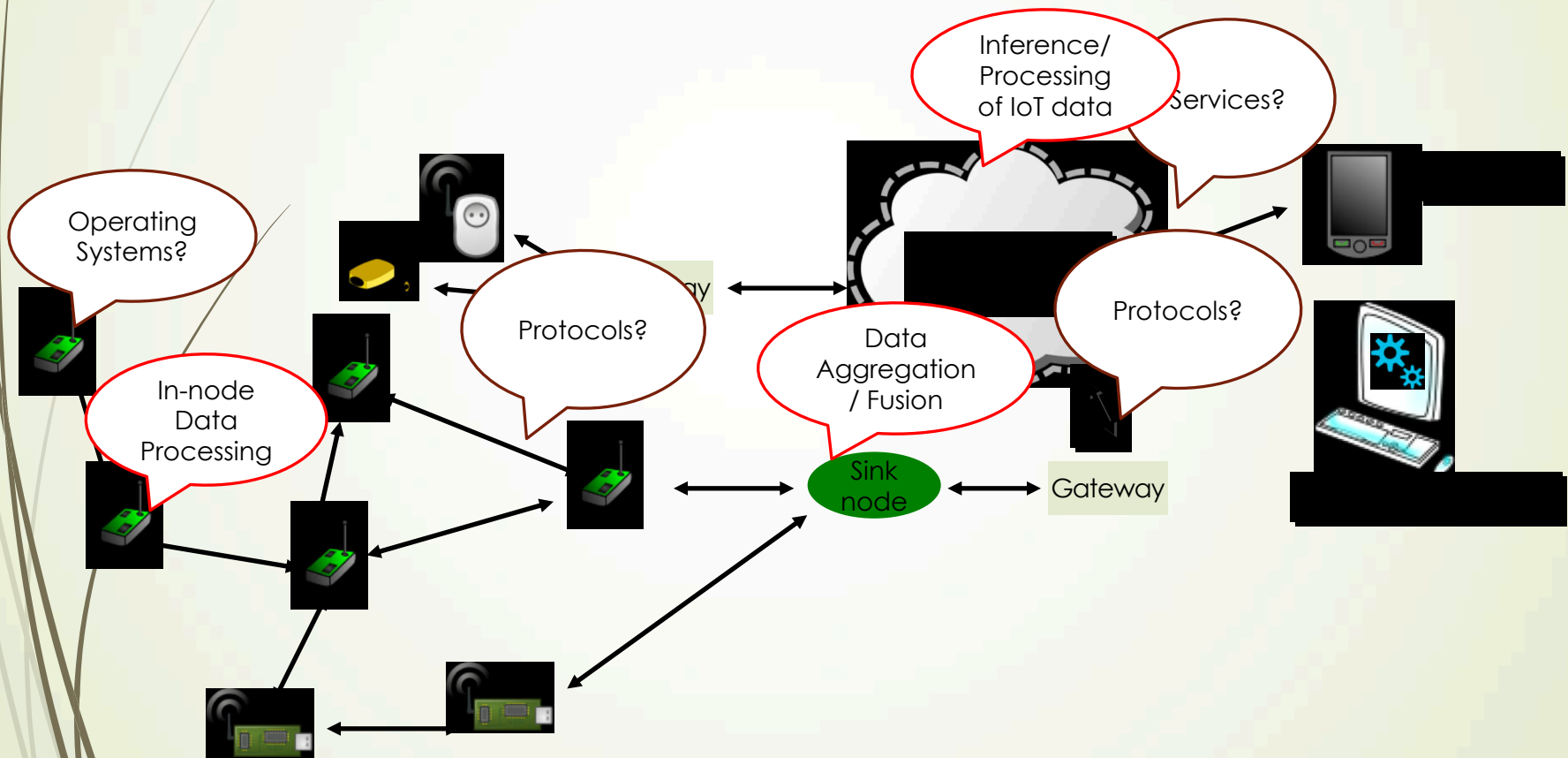
Real-Time Systems

- “In real-time computing the correctness of the system depends not only on the logical results of the computation **but also on the time at which the results are produced**” [J. Stankovic]
- Many real-time systems are control systems
- A “thing” can be modelled as a real-time system
- A multi-domain IoT application (e.g. smart city) is often a distributed real-time system
- Classification: hard/soft time constraints
- Workload characteristics: periodic/aperiodic tasks
- Scheduling:
 - Preemptive and Non-preemptive
 - Static and Dynamic
 - Online and Offline
 - Optimal and Heuristic

Agenda

- Internet-of-Things (IoT)
- IoT Versus Cyber-Physical Systems (CPS)
- IoT Data Processing
- IoT Analytics

IoT Networks



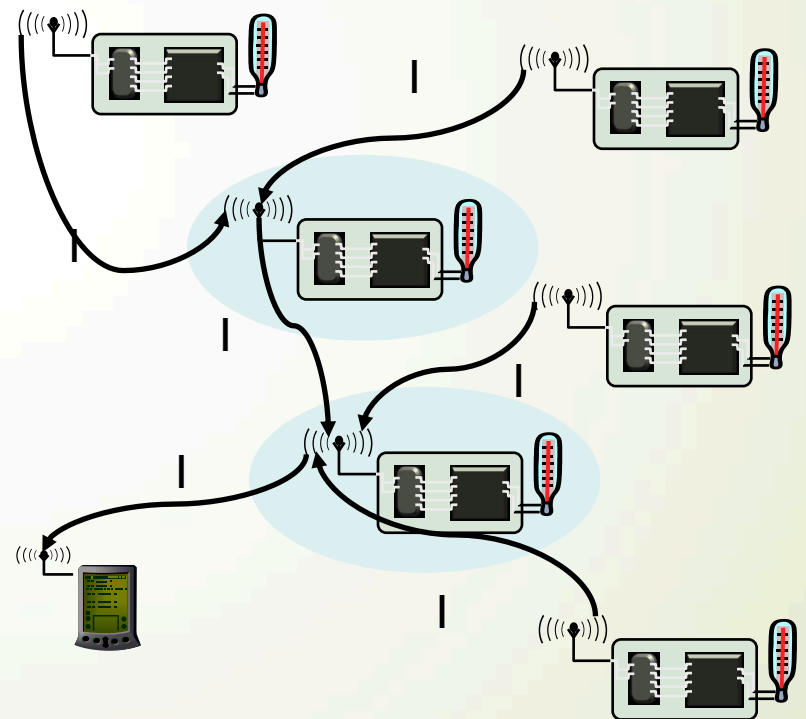
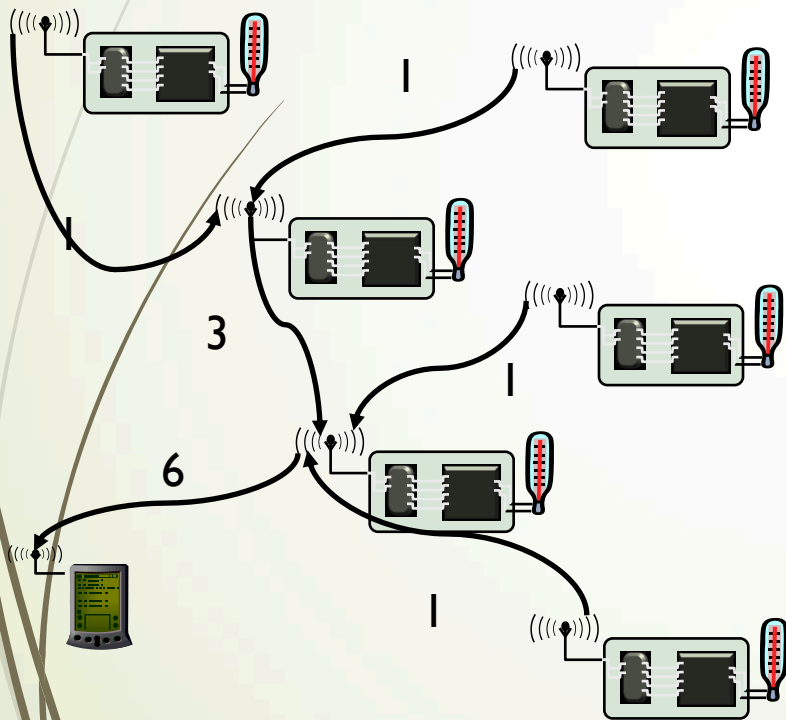
Characteristics of IoT Devices

- Often inexpensive sensors (actuators) equipped with a radio transceiver for various applications, typically low data rate $\sim 10\text{-}250$ kbps (but not always).
- Deployed in large numbers
- The sensors should coordinate to perform the desired task.
- The acquired information (periodic or event-based) is reported back to the information processing center (or some cases in-network processing is required)

In-Network Processing

- Mobile Ad-hoc Networks can be seen as a set of nodes that deliver bits from one end to the other;
- WSNs, on the other end, are expected to provide information, not necessarily original bits
 - **manipulate or process the data in the network**
- Main example: aggregation
 - Applying aggregation functions to a obtain an average value of measurement data
 - Typical functions: minimum, maximum, average, sum, ...
- More sophisticated processing of data can take place within the network or at the edge
- Exploit temporal and spatial correlation

Example of Data Aggregation

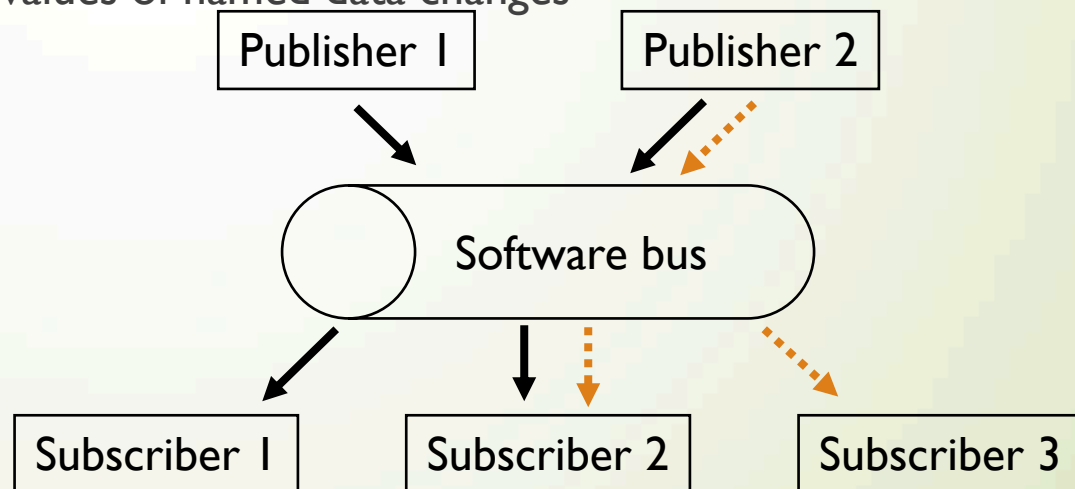


Performance Metrics for Aggregation

- Accuracy: difference between the resulting value or representation and the original data
 - Lossless versus lossy
- Coverage: the percentage of all the data items that are included in the computation of the aggregated data.
- Latency: delay time to compute and report the aggregated data
 - Computation foot-print; complexity;
- Overhead: the main advantage of the aggregation is reducing the size of the data representation;
 - Aggregation functions can trade-off between accuracy, latency and overhead;
- Aggregation should happen close to the source.

Data Exchanges

- Achieved by ***publish/subscribe*** paradigm
 - Idea: Entities can publish data under certain names
 - Entities can subscribe to updates of such ***named data***
- Conceptually: Implemented by a software bus
 - Software bus stores subscriptions, published data; names used as filters; subscribers notified when values of named data changes



MQTT Pub-Sub Protocol

- MQTT is designed to be open, simple, lightweight and easy to implement.
- A small transport overhead (**the fixed-length header is just 2 bytes**)
- It supports publish/subscribe message pattern to provide one-to-many message distribution and decoupling of applications
- The use of TCP/IP to provide basic network connectivity
- Three qualities of service for message delivery:
 - "At most once"
 - "At least once"
 - "Exactly once"

Time-Series Sensor Data

- The sensor data (or IoT data in general) can be seen as time-series data.
- A sensor stream refers to a source that provide sensor data over time.
- The data can be sampled/collected at a rate (can be also variable) and is sent as a series of values.
- Over time, there will be a large number of data items collected.
- Using time-series processing techniques can help to reduce the size of the data that is communicated